Mohammad Amaan

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Profile

Computer Science & Engineering student specializing in Data Science & AI, with a strong foundation in full-stack development (**React.js**, **Node.js**, **Express.js**) and AI-driven solutions built with **Python** and modern ML frameworks. Focused on designing intuitive, scalable applications that apply AI to real-world problems.

Work Experience

One Big Bit — **Integral University**
Backend & API Developer — Web Development — *2026*

- Engineered **RESTful APIs** using **Node.js** and **Express** for an AI-powered Hospital Management System, and designed **MongoDB** schemas, routes, controllers, and service-layer components for core operations.
- Integrated the **Google Gemini API** to build an AI-powered chatbot, adding generative-AI capabilities to the healthcare application.
- Collaborated within a **6-member** team to integrate backend services with the frontend and deliver core application modules.

eTrainIndia — **Remote**
Data Science Intern — Global Certification + Internship Program — *2025*

- Developed an Urban Noise Pattern Analysis project with **Python**, **Pandas**, **NumPy**, **Matplotlib**, **Seaborn**, and **Scikit-learn**, processing and analyzing **57,449** records across geographic, temporal, and sound-event attributes.
- Built and evaluated **Logistic Regression** and **Random Forest** classification models using feature scaling and train-test validation.
- Awarded **"Best Intern of the Program"** for exceptional commitment, initiative, and contributions during the Global Certification + Internship Program.

Projects

AI-Powered Hospital Management System (GitHub)
React.js, Node.js, Express.js, MongoDB, Gemini AI

- Developed a secure full-stack hospital management system with role-based authentication for patients, doctors, and administrators, using **JWT** auth and **REST APIs**.
- Integrated **Gemini AI** for intelligent symptom analysis and preliminary health recommendations.
- Implemented appointment booking, doctor management, patient records, and responsive dashboards.

Explainable AI in Healthcare (GitHub)
Python, PyTorch, OpenCV, Streamlit

- Built a deep learning model to classify brain **MRI** scans into four tumor categories, integrating **Grad-CAM** to visualize the regions influencing predictions.
- Developed an interactive **Streamlit** application for prediction and explainability visualization, improving transparency of AI-assisted diagnosis.

Education

Integral University — **Lucknow, India**
B.Tech, Computer Science & Engineering — Data Science & AI (in association with IBM) — *Aug 2023 – May 2027*

- CGPA: **9.0**

Skills & Interests

Languages: Python, JavaScript, Java, C
Web Technologies: HTML, CSS, React.js, Node.js, Express.js, REST APIs
Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, OpenCV, Grad-CAM
Databases: MongoDB, MySQL
Developer Tools: Git, GitHub, VS Code, Postman, Google Colab, Jupyter Notebook, Overleaf
Cloud & AI Tools: Gemini API, ChatGPT, Perplexity
Core Concepts: Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks (CN)
Soft Skills: Problem Solving, Analytical Thinking, Communication, Team Collaboration, Adaptability

Certifications & Training

- **Programming in Java** — NPTEL
- **Statistics 101** — IBM
- **Data Visualization with Python** — IBM
- **Data Science 101** — IBM
- **Data Science Tools** — IBM
- **Build Your Own Chatbot** — IBM
- **Cloud Application Developer** — IBM

Languages

- **English** (Proficient)
- **Hindi** (Native)
- **Urdu** (Native)